Raphaël Honigsberg

PhD candidate ; Microfluidics, microfabrication & microbial imaging - ENS Paris-Saclay

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Seeking a postdoctoral position from January-February 2027 - microfluidics, 3D microfabrication, single-cell imaging. Academic, open to industry.

PROFILE

Biologist trained in microfluidics. I design and build microfluidic platforms end to end (CAD, 3D-printed molds, clean-room processes, live imaging and automated image analysis) to follow microbial growth down to single cells.

Next goal: a **scalable fabrication strategy for functional 3D microfluidics**, in an academic lab or with industry.

RESEARCH EXPERIENCE

2023 - Jan. 2027 — **PhD candidate - LBPA & LuMIn, ENS Paris-Saclay, CNRS, Université Paris-Saclay**

Supervisors: E. Deprez, B. Le Pioufle, S. Bensalem ; Competitive doctoral contract (CDSN)

- *Thesis:* development of a microfluidic system to study microalgae-bacteria interactions, to optimise biofuel production yields
- Designed a pump-free, pipette-operated PDMS chip with **400 gravity-trap wells**, cast from two 3D-printed molds, one for the wells and one for the channels (no clean room)
- Built on-chip **light × chemical gradient screening**: 100 conditions × 4 replicates per chip, up to 9 chips per run
- Multi-day time-lapse epifluorescence and confocal imaging of *Chlorella vulgaris* and co-cultured bacteria, down to single cells; automated acquisition in NIS-Elements
- Image-analysis pipeline (StarDist segmentation, growth-rate fitting) on millions of segmented cells; COMSOL modelling of flow, shear and diffusion

Jan. - Jun. 2023 — **Master's thesis - Institut Pasteur (G.-F. Richard) & LadHyX, École Polytechnique (C. N. Baroud)**

- Single-cell analysis of the checkpoint response to DNA double-strand breaks in *S. cerevisiae*
- Designed a dedicated microfluidic chip; fluorescence microscopy and image analysis

Oct. 2021 - Jul. 2022 — **Pre-doctoral research year (ARPE) - Faherty Lab, Harvard Medical School & Massachusetts General Hospital, Boston**

- Adherence factors of *Shigella flexneri* (BSL-2): infection assays on HT-29 monolayers, λ-Red mutagenesis, cloning, biofilm assays - Co-author, *Gut Microbes* 2024
- Supervised two students (one graduate, one master's)

2019 - 2020 — **Research internships - Institut de Chimie Physique, UMR 8000, Université Paris-Saclay (M. Erard)**

- Fluorescent proteins for labelling cellular endomembranes: cloning, transfection, FRET-FLIM microscopy

INNOVATION & AWARDS

2025 — **Créativité Prize**, Starthèse start'up pitch-ID, Université Paris-Saclay - project *Honigs*, 3D-printed microfluidics for cell culture ; selected to pitch at **BIG 2025** (Bpifrance), Paris

2023 — Competitive doctoral contract for ENS graduates (CDSN)

TEACHING & SUPERVISION

2021 - 2022 — Supervision of two students (graduate and master's), Harvard Medical School / MGH

2 years — Oral examiner (*khôlles*) in biology, CPGE BCPST, Paris - 1 h/week ; exam grading

PUBLICATIONS & COMMUNICATIONS

León Y, **Honigsberg R**, Rasko DA, Faherty CS. Gastrointestinal signals in supplemented media reveal a role in adherence for the *Shigella flexneri* sap autotransporter gene. *Gut Microbes* 16(1), 2024. doi:10.1080/19490976.2024.2331985

Honigsberg R, et al. 3D-printed mould for a through-hole array: a pump-free chip for light × nitrate phenotyping of microalgae (working title). In preparation for *Lab on a Chip*.

Honigsberg R, Ghasemi R, Audibert JF, Deprez E, Le Pioufle B, Bensalem S. Low-cost microfluidic platform to monitor early growth from single microalgal cells in bacteria co-culture. **Poster, MicroTAS 2026**, Granada, Spain.

EDUCATION

2023 - 2027	**PhD**, Université Paris-Saclay, doctoral school SDSV (ED 577) - defence January 2027
2019 - 2023	**École Normale Supérieure Paris-Saclay** - normalien (civil-servant student), ENS diploma 2023
2022 - 2023	**MSc** Fundamental Microbiology, Université Paris Cité & Institut Pasteur
2020 - 2021	**MSc year 1** Biology & Health (Boris Ephrussi track), Université Paris-Saclay, with honours
2019 - 2020	**BSc** Biology & Health, Université Paris-Saclay, with honours
2016 - 2019	**Classes préparatoires** BCPST, Lycées Henri-IV and Fénelon, Paris - admitted to ENS Paris-Saclay (top 2 %)

SKILLS

Microfabrication	CAD at expert level: Fusion 360 (molds, chips), L-Edit (mask layout) ; 3D printing (SLA, DLP) of molds and devices ; SU-8 photolithography, single- and multi-layer (Süss MJB4, RC8 spin coater, fully autonomous) ; Dektak profilometry ; PDMS soft lithography ; plasma bonding (Diener Atto/Femto) ; UV-ozone ; parylene deposition (SCS PDS 2010, full mechanical maintenance)
Materials & surfaces	Hydrogels for microfluidics: alginate blended with CMC, methylcellulose, xanthan or gum arabic (rheology), PEGDA ; gel-surface interactions and meniscus pinning to hold gel walls in channels ; silane chemistry and surface coatings (silanisation, parylene)
Modelling	COMSOL Multiphysics: laminar flow, shear stress, diffusion
Imaging	Epifluorescence and confocal microscopy ; NIS-Elements, automated multi-day acquisitions ; single-cell image analysis: Fiji, StarDist, Cellpose, ilastik, TrackMate
Data & AI	Python analysis pipelines (images, growth fits, statistics, PCA) built with AI coding assistants ; AI-assisted literature screening and task automation ; publication-quality figures
Biology	Bacteria (*E. coli*, *S. flexneri* BSL-2, *P. stutzeri*; wild marine bacteria isolated from oyster and periwinkle digestive tracts, screened for cellulolytic strains degrading microalgae), yeast, microalgae (*Chlorella*, *Nannochloropsis*; synchronised cultures), algae-bacteria co-cultures, human cell lines (HT-29) ; molecular biology: cloning, PCR, λ-Red mutagenesis
Languages	French (native) ; English (C1-C2, Cambridge Advanced; IELTS 8.0) ; Spanish (A2)

REFERENCES

Dr Eric Deprez
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